

Dhruv Nirmal

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Summary

Data Engineer with 2+ years delivering production-grade ETL pipelines and data-quality systems on enterprise financial datasets. Designed and deployed 20+ automated pipelines processing millions of transactions monthly, achieving ~15% client savings through anomaly detection on procurement/financial spend data. Skilled in Python, SQL, Azure cloud infrastructure, and end-to-end pipeline ownership. Proven ability to reverse-engineer undocumented systems and take autonomous ownership of data platform work. Currently expanding Airflow DAG orchestration skills to complement existing pipeline automation expertise.

Experience

Data Engineer — Purchasing Index Data Analytics (Comprara Group) Jun 2024 – Present

Melbourne, Australia

- Built and maintained 20+ automated ETL pipelines processing millions of procurement and financial transactions monthly across 5 enterprise clients; owned end-to-end pipeline lifecycle from ingestion through transformation to reporting delivery (Python, SQL, Azure-hosted servers).
- Performed anomaly detection on financial procurement datasets, identifying cost-leakage patterns contributing to ~15% client savings — applied hypothesis-driven investigation on ambiguous spend signals.
- Built a multi-agent data QA validation workflow (5 agents + orchestrator) to validate client datasets prior to delivery; enforced data-quality checks at every pipeline layer before stakeholder-facing output.
- Developed 16+ Tableau dashboards for procurement and finance teams to monitor supplier spend, track financial KPIs, and identify cost-optimisation opportunities; communicated findings to senior executives.
- Automated client reporting workflows (Python, SQL, KNIME), integrating LLM tools to accelerate analytics-script development and standardise repeatable delivery processes.
- Built a binary classification model (TF-IDF + Logistic Regression) to categorise financial transaction descriptions, cutting manual categorisation effort ~40% and improving data standardisation for downstream analytics.
- Built an ML model to predict chemical consumption for industrial clients, improving production planning and inventory management; applied forecasting techniques to enterprise operational data.

Data Engineer Intern — Victorian Centre for Data Insights (VCDI) Aug 2023 – Nov 2023

Melbourne, Australia

- Built a distributed anomaly-detection pipeline in Databricks using PySpark to identify data anomalies in government procurement data; improved detection accuracy ~20%.
- Engineered scalable transformation layers to process large-scale procurement datasets in cloud environments — responsible for staging, cleansing, and mart-layer delivery.
- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders; presented data architecture and model outcomes to cross-functional teams.

Research Data Analyst — Terminal Ballistics Research Laboratory (TBRL, DRDO) Jan 2021 – Jul 2021

- Analysed blast-wave data using statistical models to predict noise levels; reduced noise ~20% via Butterworth filter; improved prediction accuracy ~20% and refined defence equipment designs ~10%.

Projects

Cloud Data Platform with Terraform (Personal)

- Provisioned full cloud infrastructure using Terraform IaC (AWS S3, IAM, EC2, networking) for reproducible multi-environment deployments (dev/staging/prod).
- Automated data ingestion pipeline from external APIs to Snowflake on Terraform-managed infrastructure; end-to-end ingestion-to-BI platform with Power BI reporting layer.

Education

Master of Data Science — Monash University Feb 2022 – Dec 2023

Statistics, Machine Learning, Applied Forecasting, Supply Chain Management, High Dimensional Analysis, Web App Development

Bachelor of Engineering — Thapar University May 2017 – May 2021

Skills

Data Engineering & SQL: Python, SQL (CTEs, window functions, stored procedures, complex queries), ETL pipeline design, data modelling, data quality & validation, anomaly detection, KNIME, Snowflake, data warehouse architecture

Cloud & Infrastructure: Azure (hosted production pipelines), AWS (S3, Lambda, IAM, EC2), Terraform (IaC), Docker

Machine Learning & AI: scikit-learn, TF-IDF, Logistic Regression, forecasting models, anomaly detection, LLM/agent workflows (Claude, ChatGPT), multi-agent orchestration, NLP/sentiment analysis

BI & Visualisation: Tableau (16+ production dashboards), Power BI, data storytelling, KPI analytics

Orchestration & Automation: Batch pipeline automation, Windows Task Scheduler (scheduled DAG-equivalent workflows), Python scripting; actively upskilling: Apache Airflow